Arduino Exercises — Colour Coded

Ex 1: Red · Ex 2: Blue · Ex 3: Green · Ex 4: Orange · Ex 5: Yellow

EXERCISE 1 — RED FONT

```
EX.NO.1 const int led1=3;

const int led2 = 4;

const int led3 = 5;

const int button1 = 6;

const int button2 = 7;

int button1 State = 0;

int button2State = 0;

void setup()

{

pinMode(led1, OUTPUT);

pinMode(led2, OUTPUT);

pinMode(led3, OUTPUT);

pinMode(button1, INPUT_PULLUP);

pinMode(button2, INPUT_PULLUP);

}

void loop()

{

button1State = digitalRead(button1);
button2State= digitalRead(button2);

25

if (button1 State LOW)

{ for(int i=0;i<10;i++)

pattern1(); // First LED pattern

{

} else if (button2State = LOW)

for(int i=0;i<10;i++)

}

pattern2(); // Second LED pattern

else

} allOff(); // Turn off all LEDs when no button is pressed

}

}

}

void pattern1()

{

digitalWrite(led1, HIGH);

delay(500);
digitalWrite(led2, HIGH);
```

```
delay(500);

digitalWrite(led3, HIGH);

delay(500);

digitalWrite(led1, LOW);

delay(500);

digitalWrite(led2, LOW);

delay(500);

digitalWrite(led3, LOW);

delay(500);

void pattern2()

}

digitalWrite(led1, HIGH);

digitalWrite(led2, LOW);

digitalWrite(led3, HIGH);
delay(300);

digitalWrite(led1, LOW);

digitalWrite(led2, HIGH);

digitalWrite(led3, LOW);

delay(300);

}

void allOff()

{

digitalWrite(led1, LOW);

digitalWrite(led2, LOW);

digitalWrite(led3, LOW);

}
```

EXERCISE 2 — BLUE FONT

```
EX.NO.2
#include <Servo.h>

int p=0;

Servo myServo;

// Create a Servo object

// IR sensor digital output connected to pin 2

const int IR_SENSOR_PIN = 2;

const int SERVO_PIN = 9;

// Servo motor control pin

void setup()
{

INPUT); // Set IR sensor pin as input

pinMode(IR_SENSOR_PIN,

myServo.attach(SERVO PIN);

// Attach the servo to pin 9

myServo.write(0);

// Initialize servo to 0 degrees

// Start serial communication for debugging

Serial.begin(9600);
}

void loop()
{

{ int sensorState = digitalRead(IR_SENSOR_PIN);

// Read the state of the IR sensor

if (sensorState = HIGH) {

// If the sensor detects an object

// Rotate the servo to 90 degrees

myServo.write(90);

Serial.println("Object detected! Rotating to 90 degrees.");

delay(1000);

// Delay to hold the servo position
}
else
{

myServo.write(0);

// Rotate the servo back to 0 degrees

Serial.println("No object detected. Returning to 0 degrees.");

delay(1000);

 }
}
```

EXERCISE 3 — GREEN FONT

```
EX.3
#include <LiquidCrystal.h>

// Initialize the LCD pins: RS, E, D4, D5, D6, D7

LiquidCrystallcd(12, 11, 5, 4, 3, 2);

// Ultrasonic sensor pins
const int trigPin = 9;

const int echoPin = 10;

void setup()

{

lcd.begin(16, 2);

// Initialize the LCD

pinMode(trigPin, OUTPUT); // Set trigPin as output

pinMode(echoPin, INPUT); // Set echoPin as input

led.print("Distance:"); // Display static text
}
void loop()

{

long duration;

float distance;

// Send a 10-microsecond pulse to trigger the ultrasonic sensor

digitalWrite(trigPin, LOW);

delayMicroseconds(2);

digitalWrite(trigPin, HIGH);

delayMicroseconds(10);

digitalWrite(trigPin, LOW);

Read the time it takes for the echo to return

duration= pulseIn(echoPin, HIGH);

// Calculate distance (in cm)

distance = duration * 0.034/2;

Display the distance on the LCD

lcd.setCursor(0, 1);

// Move to the second line

lcd.print(" ");

// Clear previous value

led.setCursor(0, 1);

lcd.print(distance);

lcd.print(" cm");
```

EXERCISE 4 — ORANGE FONT

```
EX.4
const int irSensorPin 8; // IR sensor output pin

const int buzzer Pin 9;

// Buzzer pin

const int ledPin10;

// LED pin
void setup()
{

pinMode(irSensorPin, INPUT); // Set IR sensor pin as input

pinMode(buzzerPin, OUTPUT); // Set buzzer pin as output

pinMode(ledPin, OUTPUT);

// Set LED pin as output

Serial.begin(9600);

}

// Initialize Serial Monitor for debugging

void loop()

{

int irValue = digitalRead(irSensorPin); // Read the IR sensor value

if (irValue = HIGH)

{

// Intruder detected

digitalWrite(buzzerPin, HIGH); // Turn on buzzer

digitalWrite(ledPin, HIGH); // Turn on LED

Serial.println("Intruder detected!");//

Debug message

delay(1000);

}

else

{

// No intruder

digitalWrite(buzzerPin, LOW); // Turn off buzzer
```

EXERCISE 5 — YELLOW FONT

```
EX.5
const int IdrPin = A0; // Analog pin

const int ledPin = 9;

// Digital pin connected to the LED

int Idr Value = 0;

// Variable to store LDR readings

const int threshold = 500; // Light threshold for turning the LED ON/OFF

void setup()

{
pinMode(ledPin, OUTPUT); // Set LED pim as outpat

Serial.begin(9600); // Initialize serial monitor for debugging

}

void loop() {

Idr Value = analogRead(IdrPin); // Read LDR value (0-1023)

Serial.print("LDR Value: ");

Serial.println(IdrValue); // Display LDR value on Serial Monitor

if (ldr Value threshold)

{

// Turn on the LED in darkness

digitalWrite(ledPin, HIGH);

Serial.println("LED ON");

}

else

{

// Turn off the LED in bright light

digitalWrite(ledPin, LOW);

Serial.println("LED OFF");

}

delay(200); // Delay for stability

}
```